

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

How the Web Actually Works

Q.E.D. — Web Dev Bootcamp | Session 01



Every Web App Looks Like This

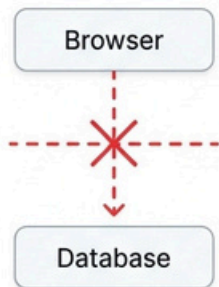
- Frontend: What the user sees. HTML, CSS, JS, React. Sends requests and displays responses.
- Backend: The brain of the app. Validates, processes, and responds via Python / FastAPI.
- Database: Permanent storage. Houses users, orders, and data safely in PostgreSQL.
- External APIs: Outsourced power. Google OAuth (Login), Emails, Payment (Stripe).



Frontend The Face of the App

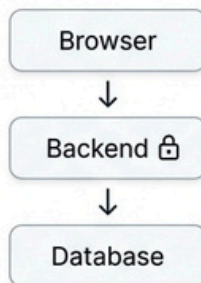
- Everything the user sees and interacts with
- Buttons, forms, pages, animations
- Built with HTML, CSS, and JavaScript
- Cannot store data or protect secrets
- It only sends requests and displays responses

Without Backend



- Credentials exposed in browser
- Anyone can access the data
- No security at all

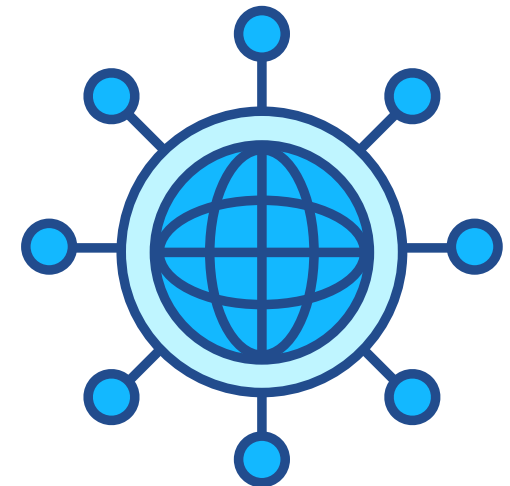
With Backend



- Code hidden from users
- Secrets stored on server
- Full validation and control

Backend Where the Real Work Happens

- Receives requests from the Frontend
- Validates — is this data correct?
- Authenticates — is this user real?
- Authorizes — is this user allowed?
- Talks to Database and External APIs
- Responds with JSON



Database — The Memory

- Stores everything permanently
- Users, orders, products, sessions
- Even if the server crashes — data survives
- Only the Backend can access it
- Requires a password (Connection String)



Without Database



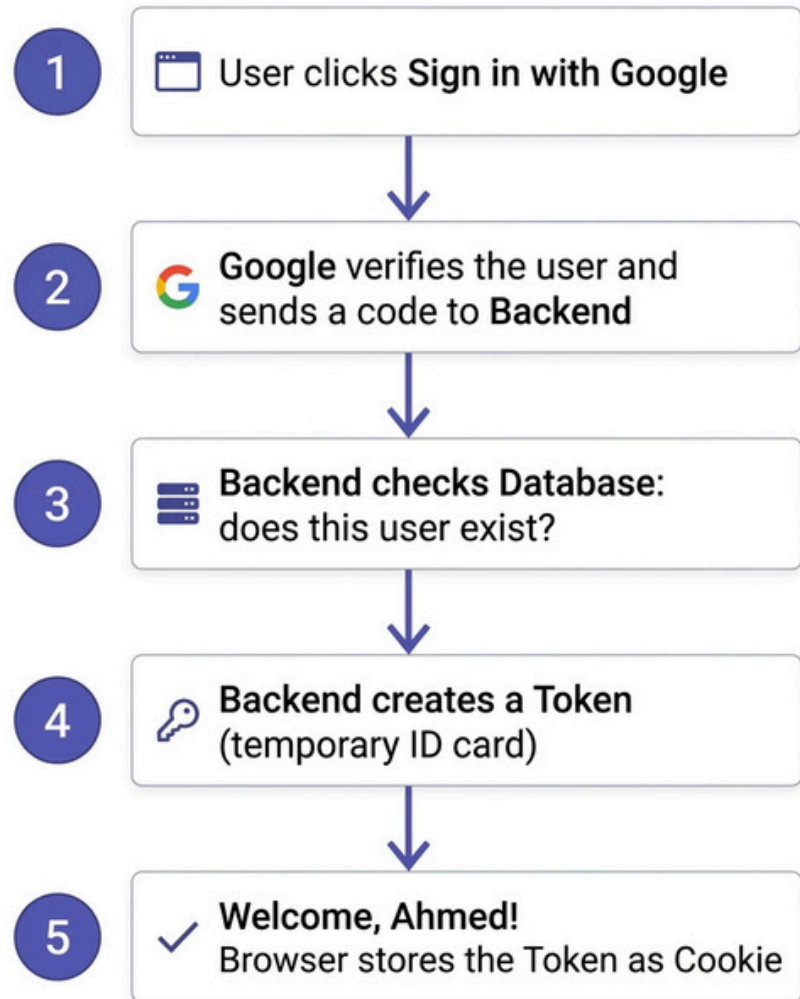
- Data stored in RAM only
- Server crashes = data lost forever
- No recovery possible

With Database



- Data stored on disk permanently
- Server crashes = data survives
- Full recovery, zero loss

Login Flow



Step 1

You click "Sign in with Google"
Browser redirects securely to Google's login page.

Step 2

Google verifies you
Sends a secret code to our Backend. Your password never touches our server.

Step 3

Backend checks the DB
"Do we have this user?"
Yes: welcome back | No: create new account.

Step 4

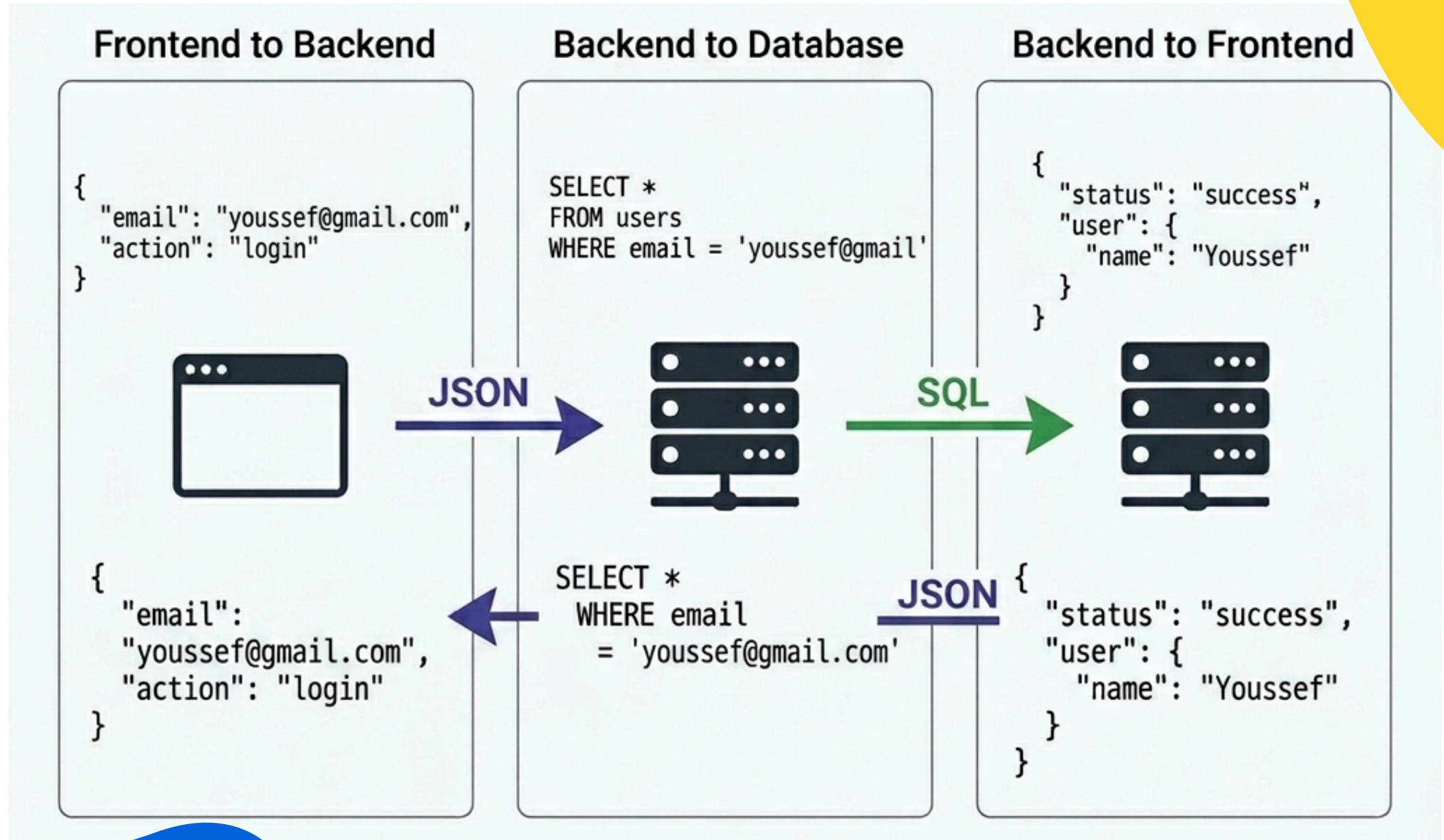
Backend creates a Token
Acts like a temporary ID card, securely stored in your browser as a Cookie.

Step 5

"Welcome, Youssef!"
Every future request includes this Token so the Backend always knows who you are.

JSON — The Universal Language

Every system in the world talks in JSON



Database Connection Strings Explained

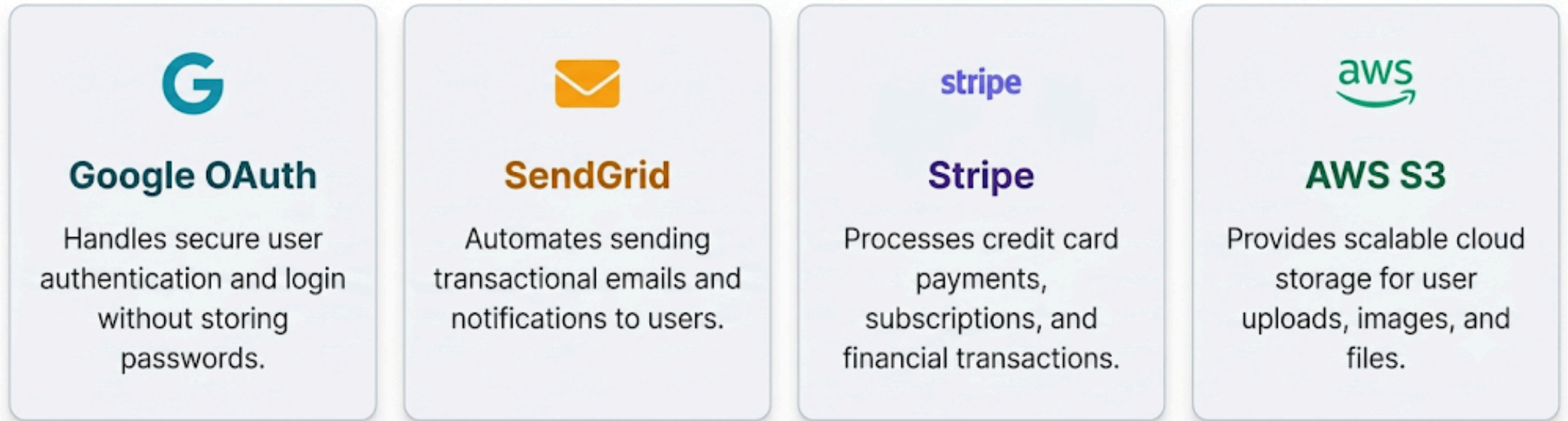
This URI is the key to your database. It contains all the necessary routing and authentication information.

```
postgresql://qed_admin:s3cur3P@ss@db.supabase.co:5432/qed_db
```



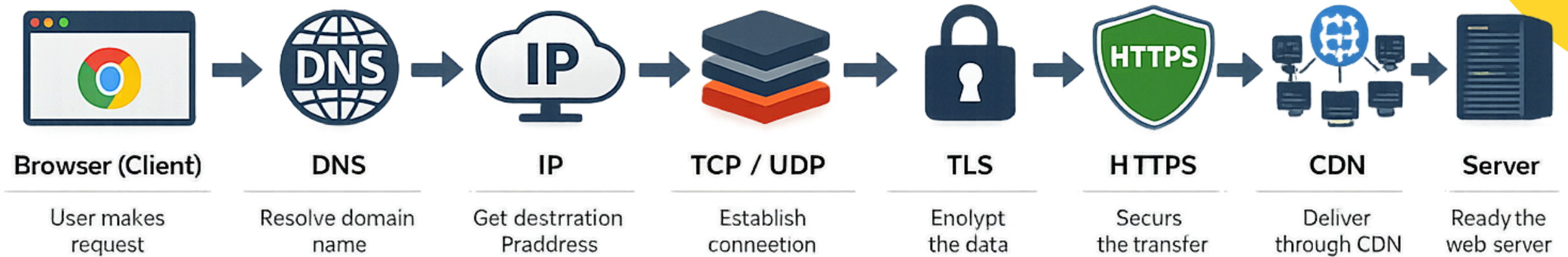
COMPONENT	VALUE
Protocol	postgresql://
Username	qed_admin
Password	s3cur3P@ss
Host	db.supabase.co
Port	5432
Database	qed_db

The Backend Talks to the Outside World



You don't build a payment system from scratch
you call Stripe. That's the power of APIs

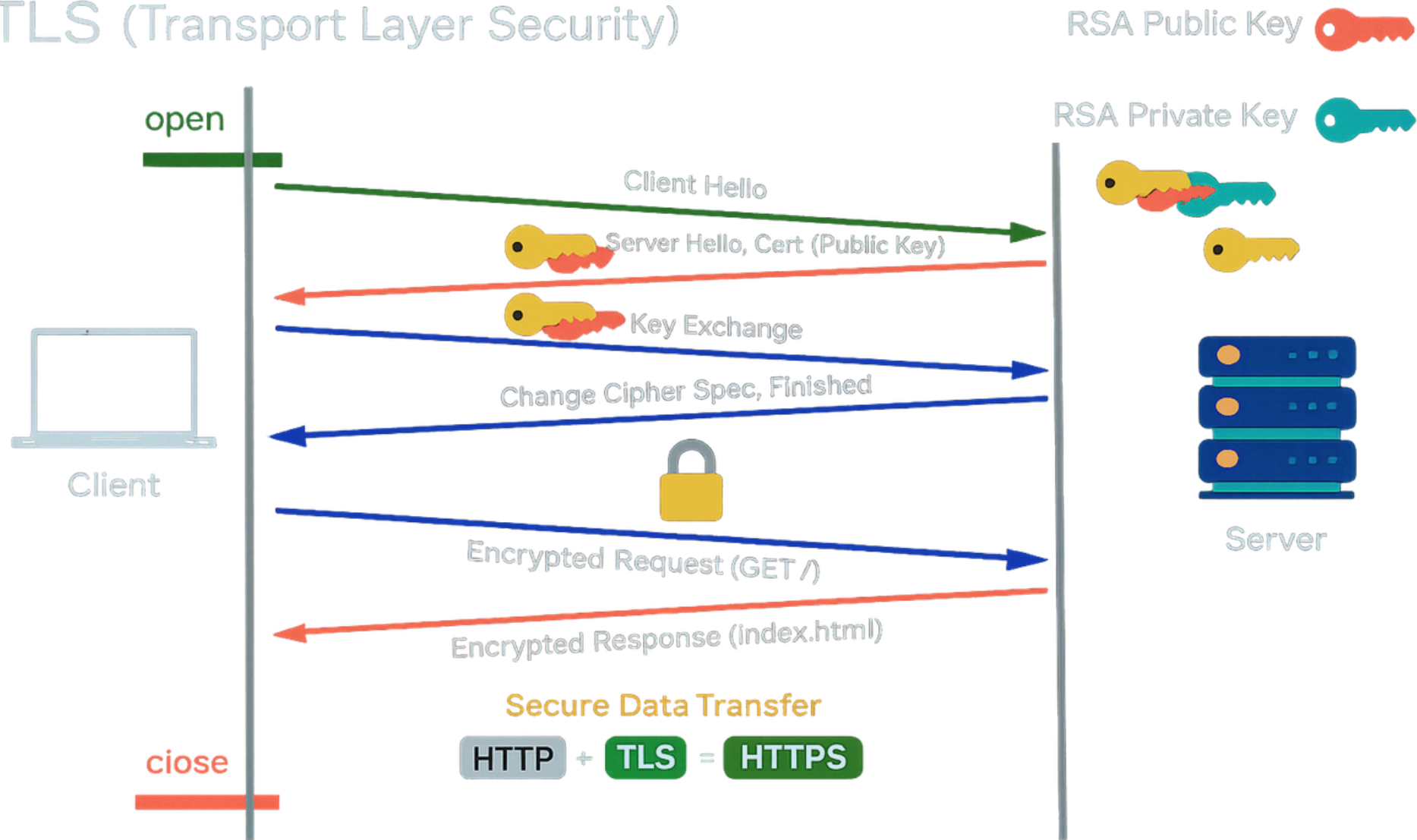
A journey with your request



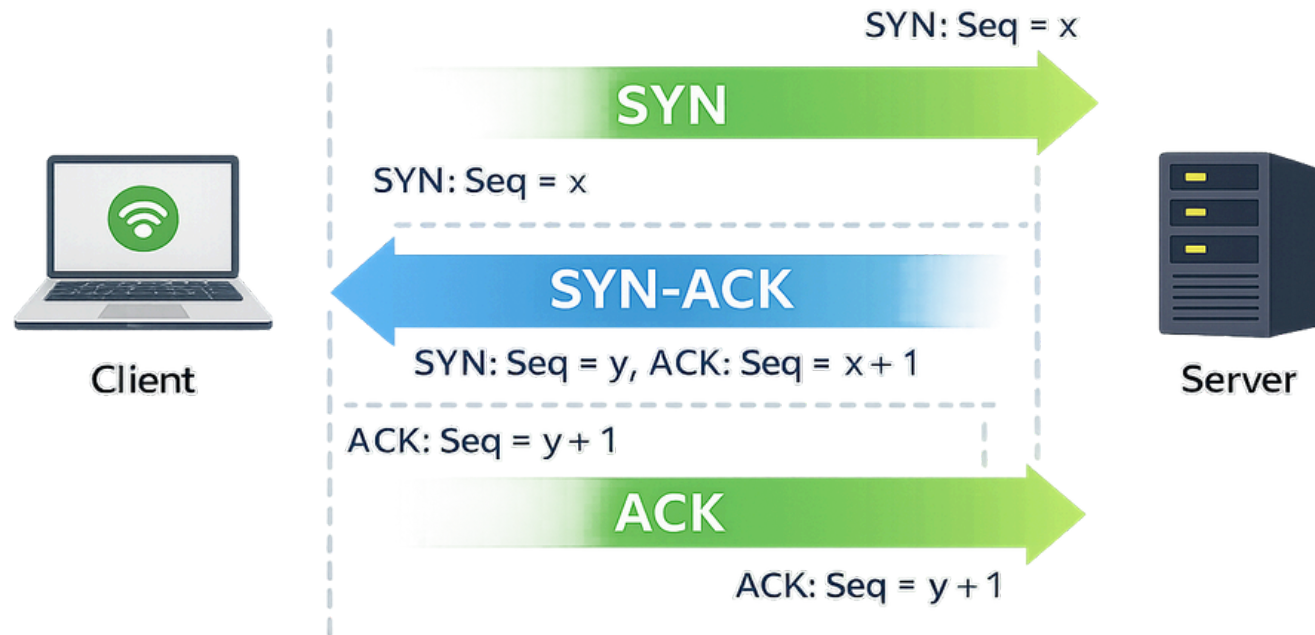
Almost every internet application communication using TCP or UDP
can be represented as:
source ip/port → destination ip/port

TLS

TLS (Transport Layer Security)



TCP HANDSHAKE



in UDP there is not any handshakes that makes it suitable when we need avoid only latency e.g. video streaming

THE BOOTCAMP

How the Bootcamp Flows



Two Ways to Learn — Only One Works

The Typical Way



Watch someone explain



Follow along passively



Get a certificate



Forget in 2 weeks

VS

The Q.E.D. Way



YOU research and prepare



YOU present and explain



YOU build a real project

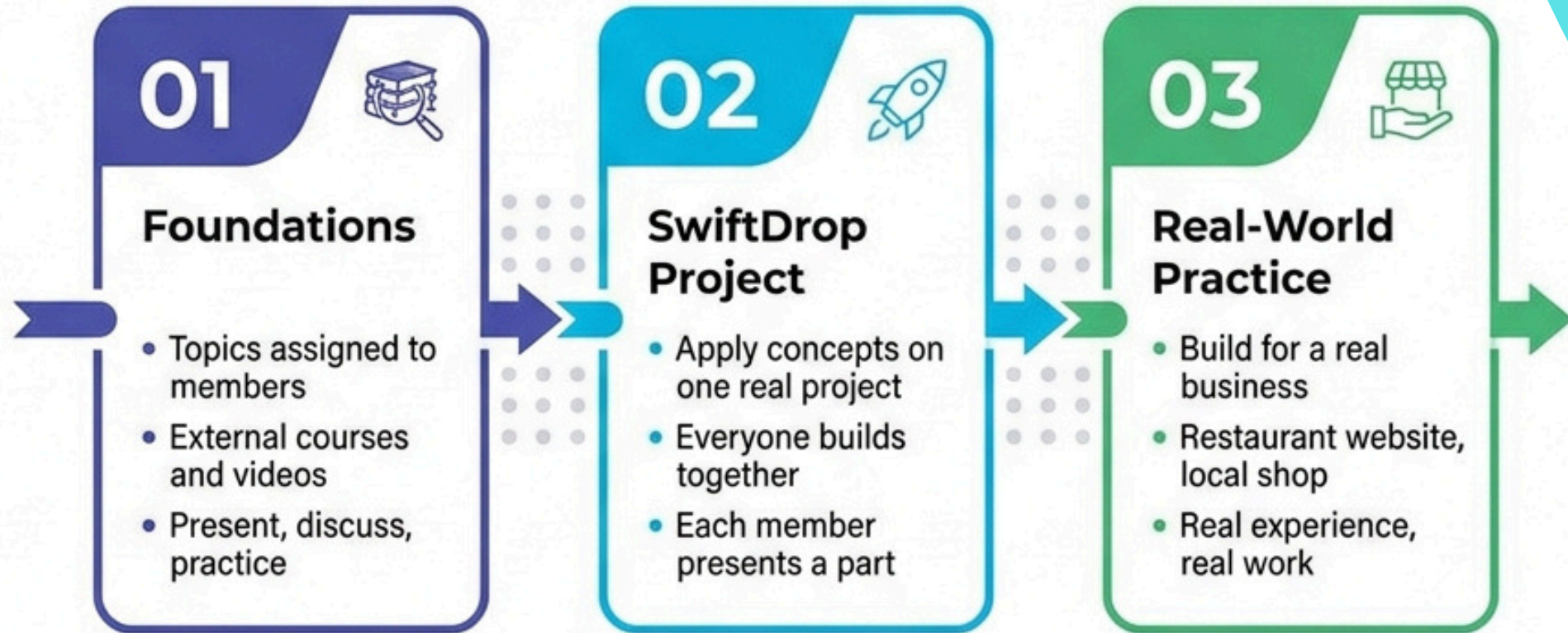


You walk away with real skills



The person who explains
learns more
than the person
who listens.

How the Bootcamp Flows



Speed doesn't matter. Understanding does. We finish when everyone is ready

What This Requires from You

Sessions will be intensive and frequent.

You need to dedicate real hours every single week.

Focus is on preparing, studying, and building — not just showing up.

If you can't commit the time, this won't be the right fit.



One Project — Every Concept Applied



Authentication & Security



Database Design & CRUD



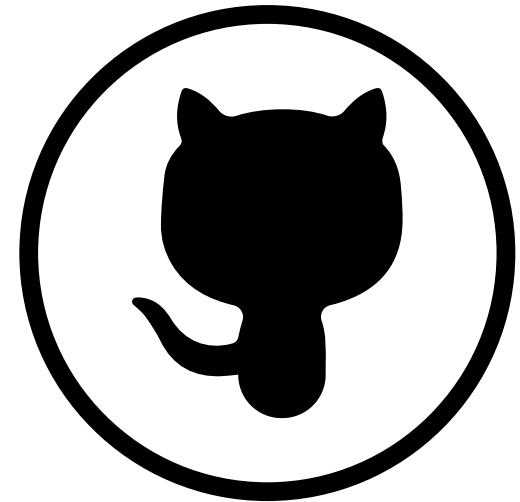
RESTful API Design



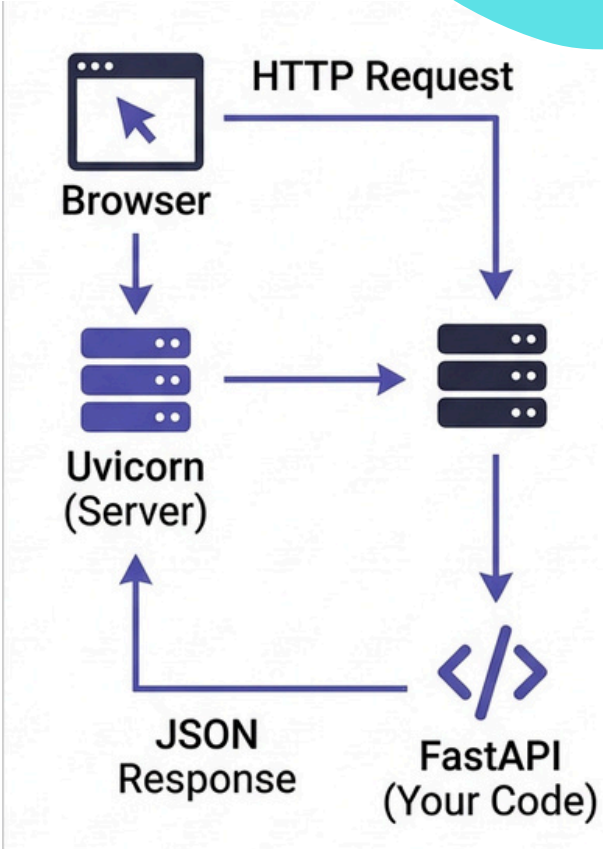
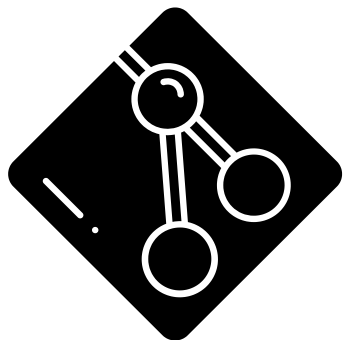
Roles & Permissions



Server Deployment



YOUR FIRST API



Setup

Install FastAPI
+ Uvicorn



Dictionary = JSON

Python dicts
become JSON



Decorators

@app.get turns
functions into
endpoints



First API

Build SwiftDrop
v0.1



/docs

Free interactive
documentation



Thank You